

6	(a)	$\frac{\text{work done or energy (transformed) (from electrical to other forms)}}{\text{charge}}$	B1	[1]
	(b)	(i) 1. $V = IR$ or $E = IR$	C1	
		$I = 14/6.0$ $= 2.3$ (2.33)A	A1	[2]
		2. total resistance of parallel resistors = 8.0Ω	C1	
		current = $14/(6.0 + 8.0)$ $= 1.0\text{A}$	A1	[2]
	(ii)	$P = EI$ (allow $P = VI$) or $P = V^2/R$ or $P = I^2R$	C1	
		change in power = $(14 \times 2.33) - (14 \times 1.0)$ or $(14^2/6.0) - (14^2/14)$ or $(2.33^2 \times 6.0) - (1.0^2 \times 14)$ $= 19\text{W}$ (18W if 2.3A used)	A1	[2]
	(c)	$I = Anvq$		
		ratio = $(0.50n/n) \times (1.8\text{A}/\text{A})$ or ratio = 0.50×1.8	C1	
		$= 0.90$	A1	[2]

Page 5	Mark Scheme	Syllabus	Paper
	Cambridge International AS/A Level – October/November 2016	9702	21